Pakistan Technology Parks Forum (PTPF) – Website Functional Requirements

1. Introduction

This document defines the comprehensive functional requirements for the **Pakistan Technology Parks Forum (PTPF)** website. The platform will serve as the digital nucleus for Pakistan's technology park ecosystem, connecting Special Technology Zones (STZs), Software Technology Parks (STPs), coworking spaces, incubators, and the companies operating within them. All features are designed to support PTPF's five core objectives: national synergy, international visibility, policy advocacy, knowledge sharing, and startup innovation.

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2. User Roles & Permissions

Role	Description	Access Level
Visitor	General public, potential members	Public pages only
Registered User	Individual (freelancer, entrepreneur) not affiliated with a member organization	Limited member directory, public resources, event registration
Company Member	Representative of a technology company (tenant)	Full member features, company profile management
Park Member	Representative of a member park (STZ, STP, coworking, incubator)	Full member features + park profile management, facility listings, tenant management
Chapter Leader	Regional chapter coordinator	Chapter page management, regional event creation, local member moderation
SIG Moderator	Leader of a Special Interest Group	SIG forum moderation, resource management

Admin	PTPF Secretariat staff	Full system control, all content management, user management, analytics

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3. Public Website (Non-Authenticated)

3.1 Home Page

Feature	Description
Hero Section	Dynamic banner with tagline: "Pakistan's Unified Voice for Technology Parks" and rotating background images of member parks
Live Statistics Counter	Real-time stats: number of member parks, total companies, provinces covered, jobs created (pull from database)
Objectives Overview	Icons linking to 5 core objectives with brief descriptions
Featured Members	Carousel showcasing founding/full member park logos with links to profiles
Upcoming Events	3-4 upcoming events with dates and brief descriptions
Latest News	Recent articles/press releases with publication dates
Success Stories	Rotating testimonials from park operators and tenant companies
Newsletter Signup	Email capture for public updates
Quick Actions	Prominent buttons: "Join PTPF", "Explore Members", "Facility Marketplace"

3.2 About Us

Feature	Description
Mission Statement	Clear articulation of PTPF's purpose and legal status (Section 42 non-profit)
Founding Secretariat	Dedicated section about Silicon Village/FDH Technology Park with stats (498 acres, STZA-licensed) and message from founding team
Objectives (Expanded)	Each of the 5 objectives presented with visuals, case examples, and relevant statistics

Governance Structure	Organizational chart (placeholder until board formation), committee listings
Annual Reports	Downloadable PDFs of annual reports (public versions)
Timeline	Interactive timeline of PTPF's establishment and key milestones
Team	Profiles of secretariat staff and board members (as appointed)

3.3 Members Directory (Public View)

Feature	Description
Search & Filters	Filter by: member type (STZ, STP, Coworking, Incubator), province/city, membership tier (Founding, Full, Associate, Affiliate)
Map View	Interactive Pakistan map with pins showing member park locations (click pin park name and type)
Grid/List Toggle	Switch between visual card grid and detailed list view
Member Cards	Each shows: logo, name, type, location, brief description, "View Profile" button
Public Profile (Limited)	Park name, logo, type, location, year established, list of facilities (summary), number of tenant companies (count only), contact form (instead of direct email)
"Become a Member" CTA	Prominent on directory pages

3.4 Companies Directory (Public View)

Feature	Description
Search	By company name, category (e.g., Fintech, AI, Healthtech), or park affiliation
Company Cards	Name, logo, category, park name, "View Profile"
Public Company Profile	Name, logo, website, description, category, year founded, park affiliation (link to park profile) – contact info hidden

3.5 Events

Feature	Description
Calendar View	Monthly grid with event indicators; click date to see events
List View	Chronological list of upcoming events with filters (type, month, location)

Event Detail	Title, date/time, location (venue/link), description, speaker/facilitator bios, agenda (if available), registration button
Registration	Public events: online form with optional payment integration; member-only events: login required
Past Events	Archive with recordings/slides (if public)
iCal/Google Calendar	One-click add to personal calendar

3.6 News & Insights

Feature	Description
Categories	Press Releases, Industry News, Policy Updates, Blog, Member Spotlights
Featured Articles	Editor's pick section
Article Detail	Title, author, publication date, featured image, full content, related articles, social sharing buttons
RSS Feed	XML feed for external subscriptions
Search	Full-text search across all articles

3.7 Resources (Public)

Feature	Description
Resource Categories	STZA Act 2021, Policy Briefs, Infographics, Sector Overviews
Downloadable Content	PDFs, presentations, videos with brief descriptions
"Member-Only" Teaser	For gated resources: show preview with login prompt

3.8 International Engagement

Feature	Description
Global Partnerships	List of international associations (IASP, etc.) and MOUs
FDI Attraction	Overview of Pakistan's tech incentives with downloadable investor deck
Delegation Visits	Photo gallery and reports from international missions
Upcoming International Events	Conferences where PTPF will participate

3.9 Membership Information

Feature	Description
Tiers & Benefits	Comparison table of Founding, Full, Associate, Affiliate memberships
Fee Structure	Clear pricing (if applicable) or "Contact for details"
Join Now Form	Multi-step form: organization details, contact person, membership tier interest, message
FAQ	Common questions about membership process and benefits

3.10 Contact

Feature	Description
Inquiry Form	Fields: name, email, phone, organization, inquiry type (Membership, General, Media, Partnership), message
Secretariat Address	Physical address with map embed (Google Maps)
Contact Details	Phone, email, social media links
Office Hours	Business hours for secretariat
Live Chat	Optional integration for immediate queries (e.g., Tidio)

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4. Member Authentication & Account Management

4.1 Registration

Feature	Description
Account Types	Individual, Company Representative, Park Representative (select during signup)
Registration Form	Email, password, organization name (if applicable), phone, verification code (email sent)
Email Verification	Link valid for 24 hours
Admin Approval	New member accounts require admin verification (for organization accounts)
Welcome Email	Contains login instructions and onboarding resources
Social Login	Optional: LinkedIn, Google for faster signup

4.2 Login

Feature	Description
Credentials	Email + password
Remember Me	Persistent login (30 days)
Forgot Password	Email reset link with expiry
Two-Factor Authentication	Optional for enhanced security (SMS or authenticator app)
Session Management	View active sessions, logout from other devices

4.3 Member Dashboard

Feature	Description
Overview Widgets	Profile completeness, unread messages, pending inquiries, upcoming events, recent forum activity
Quick Actions	Edit profile, post facility, create event (if permitted), message members
Notifications Feed	Real-time updates (messages, mentions, event reminders)
Activity Log	Your recent actions (profile updates, facility posts, forum comments)

4.4 Profile Management

Feature	Description
Organization Details	Name, logo, type, description, year established, website, social links
Contact Information	Email, phone, address (visibility settings: public, members-only, hidden)
Facilities & Amenities	List of available facilities (for parks) – can be tagged and updated
Team Members	Add/remove additional representatives for the same organization (with role-based permissions)
Profile Completeness Meter	Encourages filling all fields for better visibility
Privacy Settings	Control what information is visible to public vs. members-only

Change Password	With current password verification
Account Deletion	Self-service with admin notification (30-day grace period)

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5. Member Communication & Collaboration Tools

5.1 Member Directory (Full Access)

Feature	Description
Advanced Search	Filter by: member type, location, expertise/industry, membership tier, facilities offered
Save Searches	Save frequent filter combinations
Export List	Download selected members as CSV (admin only)
Member Badges	Visual indicators for Founding Members, Verified Parks, etc.
Connection Requests	Follow/unfollow members (mutual consent not required for following, but required for messaging if privacy set to "connections only")
View Full Profiles	All contact details visible (based on member's privacy settings)

5.2 Private Messaging

Feature	Description
Inbox	Threaded conversations with unread indicators
Compose	Send to individual members or groups (select multiple recipients)
Attachments	Send files (images, PDFs) up to 10MB
Read Receipts	See when message is read (optional, user can disable)
Message Search	Search within conversations
Block/Report	Block specific members; report inappropriate messages to admin
Templates	Save common message templates (e.g., introduction, collaboration proposal)
Email Notifications	Instant or daily digest (user preference)

5.3 Discussion Forums

Feature	Description
Forum Categories	Policy & Advocacy, Park Operations, Technology & Innovation, Startup Support, International Opportunities, General Discussion
Sub-Forums	Nested categories (e.g., "Park Operations" "Security", "Maintenance", "Tenant Relations")
Thread Creation	Rich text editor with image/video embedding, polls, file attachments
Thread Features	Subscribe, bookmark, share, report
Replies	Nested comments, upvote/downvote, best answer marking (for Q&A)
Moderation	Admin and category moderators can edit/delete posts, warn users
User Reputation	Points based on helpful posts (optional gamification)
Search	Full-text search across all forums

5.4 Special Interest Groups (SIGs)

Feature	Description
SIG Creation	Members can propose new SIG; admin approval required
SIG Types	Topic-based (e.g., "AI Park Operators"), role-based (e.g., "Incubator Managers"), regional (e.g., "Karachi Chapter")
SIG Homepage	Custom banner, description, leader(s), member count
SIG Forum	Private discussion area for members of that SIG
SIG Resource Library	Shared documents, templates, recordings (SIG members only)
SIG Events	Calendar of SIG-specific meetups, webinars
SIG Membership	Join request (auto-approve or leader approval)
SIG Notifications	Digest of SIG activity

5.5 Activity Feed

Feature	Description
Global Feed	Recent activity across all members: new forum posts, facility listings, events, news
Personalized Feed	Filtered by connections and followed SIGs

Like & Comment	Engage with posts
Share	Repost to your network or specific groups
Mentions	@username to notify specific members

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6. Marketplace & Business Facilitation

6.1 Facility Marketplace

Feature	Description
List Facility	Parks can add: title, description, space type (office, lab, coworking desk, conference room), size (sq ft), monthly rate, currency, availability, amenities, photos (up to 10), contact person
Search & Filter	By location, space type, price range, size, amenities
Map View	Show facility locations on map
Facility Detail Page	Gallery, detailed specs, contact form (inquiry sent to park's dashboard and email)
Shortlist	Save interesting facilities to compare
Inquiry Management	Park dashboard shows all inquiries with status (new, replied, closed)
Analytics for Parks	Views per listing, average response time

6.2 Company Directory (Full Access)

Feature	Description
Complete Profiles	Company name, logo, description, website, category, technologies used, investment stage, funding history (optional), team size, contact email, phone, social links
Search by Skills	Find companies with specific expertise (e.g., React developers, AI researchers)
Investment Opportunities	Companies seeking funding can mark themselves; investors can filter
Connect Button	Send message directly to company representative

6.3 Partnership & Collaboration Opportunities

Feature	Description

Post Opportunity	Title, description, type (Joint Venture, Research Collaboration, Investment, Service Provider Wanted), deadline, contact
Browse Opportunities	Filter by type, industry, location
Express Interest	Click to notify poster; pre-filled message with your profile
Success Stories	Showcase collaborations that originated on the platform

6.4 Job Board

Feature	Description
Post Job	Title, description, requirements, location (remote/onsite), job type (full-time, part-time, internship), application deadline, apply via platform (upload CV) or external link
Job Search	By keyword, category, location, job type
Job Alerts	Email notifications for new jobs matching saved criteria
Company Job Dashboard	Manage postings, view applicants, update status
Admin Moderation	Review jobs before publishing

6.5 Tenders & RFPs

Feature	Description
Post Tender	For parks seeking vendors/contractors (construction, IT services, etc.)
Browse Tenders	Filter by category, closing date
Download Documents	Attach RFP documents
Bid Submission	Online form for interested vendors (optional)

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7. Knowledge & Resource Management

7.1 Resource Library (Gated)

Feature	Description
Upload Resource	Title, description, category (Reports, Best Practices, Templates, Whitepapers), tags, file upload (PDF, PPT, DOC, XLS, video),

	access level (public, members-only, SIG-only)
Browse Resources	Grid/list view with filters by category, tags, date
Search	Full-text search within documents (if text extracted)
Download Tracking	Count downloads per resource
Featured Resources	Admin picks for homepage
User Ratings	Rate resources (1-5 stars) and leave comments
Recommendations	"Other resources you might like" based on download history

7.2 Industry Analytics Dashboard

Feature	Description
Overview Stats	Total parks, companies, employment, exports (aggregated, anonymized)
Trends Over Time	Graphs showing growth in members, facility occupancy, sector-wise company distribution
Regional Breakdown	Compare provinces/cities
Benchmarking	Parks can compare their metrics (e.g., occupancy rate) to national averages (requires opt-in data sharing)
Export Reports	Download charts as images or data as CSV/PDF
Custom Date Ranges	Filter analytics by time period

7.3 Best Practices & Case Studies

Feature	Description
Categorized Guides	Park setup, tenant management, sustainability, security, etc.
Member Contributions	Members can submit case studies; admin reviews before publishing
Downloadable Formats	PDF versions for offline reading

7.4 Webinar & Training Archive

Feature	Description

Recorded Sessions	Embed videos (YouTube/Vimeo) with descriptions and slides
Certificates of Completion	For members who attended live (PDF generation)
Quiz/Assessment	Optional knowledge checks after training modules

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8. Events & Conference Management

8.1 Event Creation (Admin & Chapter Leaders)

Feature	Description
Event Details	Title, description, type (Conference, Workshop, Webinar, Networking), start/end date/time, timezone, location (physical address or virtual link), capacity, registration deadline, fee (if any), featured image
Speakers/Agenda	Add speaker names, bios, photos; create agenda with sessions
Registration Form	Custom fields (e.g., dietary preferences for physical events)
Payment Integration	Connect with Stripe/JazzCash for paid events
Email Reminders	Automatic reminders to registered attendees (24h before, 1h before)
Post-Event	Upload recording and slides; send thank-you email with survey

8.2 Event Registration (Member)

Feature	Description
Browse Events	Calendar, list, map views
Register	One-click (pre-filled from profile) or custom form
My Events	Dashboard showing upcoming and past registered events
Calendar Sync	Add to Google/Outlook/iCal
Waitlist	If event is full, join waitlist; auto-notify when spot opens
Cancel Registration	With optional reason

8.3 Event Management (Organizer Dashboard)

Feature	Description

Attendee List	View, search, export (CSV)
Check-in	QR code scanner for physical events
Send Message	Email all attendees (e.g., last-minute updates)
Feedback Survey	Send post-event survey; view results

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9. Advocacy & Policy Center

9.1 Policy Resources

Feature	Description
Current Priorities	List of active policy issues PTPF is working on
Position Papers	Downloadable PDFs with PTPF's official stance
Government Consultations	Open consultations with deadlines; members can submit input via form
Advocacy Toolkit	Templates for writing to government officials, key messages, data points

9.2 Member Action Alerts

Feature	Description
Alert Creation (Admin)	Title, description, call to action (e.g., "Sign this petition", "Submit comments by date"), target audience (all members or specific types)
Email Notification	Send alert to relevant members
Response Tracking	Monitor how many members took action
Impact Reports	Summarize outcomes of advocacy efforts

9.3 Legislative Tracker

Feature	Description
Track Bills	Monitor relevant legislation (import/export, taxation, tech policy) with status updates
Member Updates	Notify members when bills progress

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10. Regional Chapters

10.1 Chapter Homepages

Feature	Description
Chapter Leader Profiles	Contact info and message
Local Events	Filtered events for that region
Local News	Region-specific announcements
Member Directory	Filtered to show members in that province/city
Discussion Forum	Regional topics

10.2 Chapter Leader Tools

Feature	Description
Dashboard	Overview of chapter members, pending event approvals, recent activity
Create Local Event	With option to auto-notify all chapter members
Send Broadcast	Email all chapter members
Manage Chapter Page	Update banner, description, featured content

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11. Admin Panel

11.1 Dashboard

Feature	Description
Key Metrics	Total members, pending approvals, new inquiries, upcoming events, system alerts
Recent Activity	Latest member registrations, forum posts, facility listings
Quick Actions	Create user, add news, generate report

11.2 Member Management

Feature	Description
List All Members	Search, filter by type/tier/status, export

Member Details	View full profile, edit, suspend, delete
Approve New Members	Review registration details, approve/reject with optional note
Membership Tier Management	Change tier, set renewal dates
Bulk Actions	Email selected members, export data

11.3 Content Management

Feature	Description
Page Editor	WYSIWYG editor for all public pages (Home, About, etc.)
News/Articles	Create, edit, publish, feature
Resources	Upload, categorize, set access level
Events	Create, edit, cancel, view registrations
Facility Listings	Moderate new listings, feature/hide

11.4 Forum Moderation

Feature	Description
Reported Content	List of flagged posts with context
User Warnings	Issue warnings, suspend forum access
Category Management	Create/edit/delete forum categories and sub-forums
SIG Management	Approve new SIGs, assign moderators

11.5 Communications

Feature	Description
Broadcast Email	Compose and send to selected member segments; track open/click rates
Announcements	Create pop-up announcements for member dashboard
Push Notifications	Send to mobile app users (future)

11.6 Reports & Analytics

Feature	Description
Member Growth	Charts over time by type/tier

Engagement Metrics	Forum posts, messages, facility inquiries per month
Event Attendance	Reports by event, member type
Export All	Generate comprehensive CSV reports

11.7 System Settings

Feature	Description
General Settings	Site name, logo, contact info, social links
Membership Settings	Define tiers, fees, approval workflow
Email Templates	Customize all system emails (welcome, reset password, etc.)
Integrations	Payment gateway, email service, map API keys
Backup	Manual and scheduled database backups
Logs	View system logs, user activity logs

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12. Notifications & Alerts

12.1 In-App Notifications

Trigger	Notification Content
New private message	"You have a new message from [Name]"
Forum reply	"[Name] replied to your thread [Title]"
Mention	"[Name] mentioned you in [Post]"
Event reminder	"Reminder: [Event Title] starts in 2 hours"
Facility inquiry	"New inquiry about your facility [Title]"
Membership approved	"Your membership application has been approved"
Admin announcement	"[Subject] – [Brief]"
Connection request	"[Name] wants to connect"

12.2 Email Notifications

Type	Frequency	Opt-Out Option
Message received	Instant or daily digest	Yes
Forum activity	Instant or daily digest	Yes
Event reminders	24h and 1h before	No (can disable per event)
Policy alerts	As sent by admin	No (core benefit)

Newsletter	Weekly/monthly	Yes
System (password reset, etc.)	Instant	No

12.3 Notification Preferences

Feature	Description
Central Settings	Users choose which notifications they receive and how (in-app, email, both, none)
Quiet Hours	Suppress notifications during set times
Digest Options	Real-time, hourly, daily, weekly

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13. Mobile Responsiveness

Feature	Description
Responsive Design	All pages fully functional on smartphones and tablets
Touch-Friendly	Buttons and menus sized for touch
Mobile Navigation	Hamburger menu for small screens
Mobile Dashboard	Simplified view with key metrics and quick actions
Progressive Web App (PWA)	Optional: install as app on home screen with offline access to some content

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14. Technical Specifications

Area	Requirement
Frontend	React 18+ (SPA) with lazy loading
Backend	Node.js + Express REST API
Database	MongoDB with Mongoose ODM
Authentication	JWT (HttpOnly cookies) + optional 2FA
Real-time	Socket.io for messaging and notifications
File Storage	AWS S3 or Cloudinary
Search	MongoDB Atlas Search or Elasticsearch
Email	SendGrid / AWS SES / Nodemailer
Payments	Stripe / JazzCash integration

Hosting	Frontend: Vercel/Netlify; Backend: AWS/DigitalOcean; DB: MongoDB Atlas
Security	HTTPS, rate limiting, data encryption, XSS/CSRF protection
Performance	Page load < 3s, API response < 500ms
Scalability	Support 5000+ concurrent users